Early Retirement and Labour Force Participation: Evidence from Germany's 'Rente mit 63' Reform

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Abstract

Since July 2014, German workers with at least 45 contribution years can claim a full pension at age 63 without actuarial deductions. We estimate the effect of this reform on the employment of eligible men using SOEP data and a difference-in-differences design that compares men aged 62 to 65 above the eligibility threshold to a control group with 42 to 44 contribution years over the period 2009 to 2019. Our preferred individual fixed effects specification yields a treatment effect of -0.219 ($p < 0.001$), corresponding to a 22 percentage point reduction in the probability of employment. Neither region nor educational attainment significantly moderates the effect. The income distribution, however, reveals a steep gradient: the lowest quintile reduces employment by 35.1 percentage points ($p < 0.001$), while the highest quintile shows no statistically significant response (-4.6 pp, $p = 0.392$). The result survives alternative control group definitions, placebo tests, and a donut design. Taken together, the findings document a substantial labour market withdrawal concentrated among lower earners, with fiscal costs distributed across all remaining contributors.

Author Contributions

This paper was written jointly by Konstantin Böhm and Anton Wilhelmi. The contributions of each author are as follows:

Konstantin Böhm is responsible for the Abstract and Sections 1 (Introduction), 2 (Institutional Background), 3 (Data & Empirical Strategy), and 7 (Conclusion).

Anton Wilhelmi is responsible for Sections 4 (Results), 5 (Robustness Checks), and 6 (Discussion).

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1 Introduction

Pay-as-you-go pension systems rest on an implicit contract between generations: current workers finance current retirees' pensions, trusting that the next generation will do the same for them. The viability of this contract depends on a stable ratio of contributors to beneficiaries. When a government enacts a reform that simultaneously reduces the number of contributors and extends the duration of benefit receipt, it does not create new resources; it shifts existing ones. The costs are borne by those who continue to pay in.

Germany's *Rente mit 63*, effective from July 2014, granted workers with at least 45 qualifying contribution years the right to claim a full pension at age 63 without actuarial deductions, removing the financial penalty that had previously discouraged early labour market exit. Within its first three years, the reform generated direct fiscal costs of 6.5 billion euros for the statutory pension insurance; when foregone tax revenues and social contributions are included, the total burden reached an estimated 12.5 billion euros, roughly 30 percent above the government's own projections (Börsch-Supan et al. 2015). These are not investment expenditures that expand future productive capacity. They are consumption transfers to current retirees, financed through current and future contribution rates.

The demographic context in which these transfers take place lends them particular weight. Germany's natural population deficit, defined as the excess of deaths over births, has widened steadily over the past two decades and reached an estimated 352,000 in 2025, the largest post-war shortfall on record, driven by the lowest birth figure since 1946 (see Figure A.1 in Appendix A).[1] The old-age dependency ratio, measuring the number of persons aged 65 and older per 100 persons of working age, stood at 39 in 2024, up from 16 in 1950, and is projected to rise to between 43 and 61 by 2070, depending on assumptions about fertility, life expectancy, and net migration (Statistisches Bundesamt 2025) (see Figure A.2 in Appendix A). *Rente mit 63* was enacted at the onset of the steepest projected phase of demographic ageing in post-war German history, precisely the moment at which every additional year of labour force participation is most valuable for the sustainability of the pension system. A reform that induces large-scale early exit under these conditions does not merely shift costs forward; it compounds a structural imbalance that is already accelerating.

The scale of these fiscal commitments raises a prior empirical question: how large was the behavioural response? If eligible workers barely changed their retirement behaviour, the costs would remain bounded by the mechanical effect of removing deductions. If, however, the reform induced a large-scale withdrawal from employment, the fiscal implications compound: each worker who exits the labour force earlier simultaneously reduces tax revenue and social insurance contributions while drawing

[1] Figures for 2024 and 2025 are preliminary (Statistisches Bundesamt 2026).

benefits for longer. The most direct evidence to date comes from Dolls and Krolage (2023), who use administrative pension records and a bunching design to show that eligible men claim their pensions on average nine months earlier than comparable non-eligible workers. Their analysis focuses on the timing of pension claiming, however, and does not examine whether the reform also reduced actual employment, nor whether the behavioural response differed across the income distribution, a margin with direct implications for who bears the costs of the reform and who reaps its benefits.

This paper addresses both questions. Using data from the Socio-Economic Panel (SOEP) and a difference-in-differences framework, we compare men aged 62 to 65 with at least 45 contribution years to a closely matched control group with 42 to 44 contribution years, who faced identical labour market conditions but were not eligible for penalty-free early retirement. Our preferred specification with individual fixed effects yields a DiD estimate of -0.219 ($p < 0.001$), implying a reduction in the probability of employment of approximately 22 percentage points among eligible men. The effect is robust across alternative control group definitions, placebo tests in false reform years, and a donut design excluding workers near the eligibility threshold. We find no significant heterogeneity by region or educational attainment. The income distribution, however, reveals a pronounced gradient: the employment decline is largest among the lowest earners and statistically indistinguishable from zero in the top quintile, consistent with a reform whose costs are socialised across all contributors while its benefits accrue disproportionately to those who could not have afforded early retirement without it.

The remainder of this paper proceeds as follows. Section 2 describes the institutional setting. Section 3 presents the data and empirical strategy. Section 4 reports the main results and heterogeneity analysis. Section 5 discusses robustness checks. Section 6 interprets the findings and acknowledges limitations. Section 7 concludes.

2 Institutional Background

Germany's statutory pension insurance (*Gesetzliche Rentenversicherung*, GRV) operates as a pay-as-you-go system in which current contributions finance current benefits. Pension entitlements are calculated on the basis of *Entgeltpunkte* (earnings points), which accumulate proportionally to annual earnings relative to the national average wage. The standard retirement age (*Regelaltersgrenze*) stood at 65 years for cohorts born before 1947 and has been gradually increasing to 67 for cohorts born in 1964 or later. Workers who claim benefits before reaching their statutory retirement age face permanent actuarial deductions of 0.3 percent per month of early claiming, a penalty designed to discourage premature labour market exit.

Prior to 2014, the primary pathway to early retirement without deductions required reaching the standard retirement age. Workers who wished to retire earlier could do so through several channels, including the *Altersrente für langjährig Versicherte* (pension for long-term insured, available from age 63 with 35 contribution years, subject to deductions) or through disability and unemployment pathways. Successive reforms in the 1990s and 2000s had progressively closed or restricted these early exit routes, raising effective retirement ages and increasing labour force participation among older workers (Riphahn and Schrader 2023).

The *Rente mit 63*, enacted through the *Rentenversicherungs Leistungsverbesserungsgesetz* and effective from 1 July 2014, reversed this trajectory for a specific group. The reform created a new early retirement pathway (*Altersrente für besonders langjährig Versicherte*) that allowed workers with at least 45 qualifying contribution years (*Wartezeit von 45 Jahren*) to claim a full, unreduced pension before the standard retirement age. For the first eligible cohort (born in 1951 or earlier), the minimum claiming age was set at 63. For subsequent cohorts born from 1953 onwards, the minimum age increases by two months per birth year, converging to 65 for cohorts born in 1964 or later.

The 45-year qualifying period counts not only periods of employment subject to compulsory contributions but also certain non-employment spells, including periods of child-rearing, caregiving, compulsory military or civilian service, and receipt of short-term unemployment benefits (Arbeitslosengeld I). Periods of long-term unemployment benefit receipt (Arbeitslosengeld II) are explicitly excluded, a provision intended to prevent strategic accumulation of contribution years through the welfare system. Börsch-Supan et al. (2015) estimate that approximately 240,000 workers per cohort met the eligibility criteria at the time of introduction, the vast majority of them men with long, uninterrupted employment biographies in manual occupations.

The fiscal implications were substantial from the outset. Within three years, the reform generated direct costs of approximately 6.5 billion euros for the GRV; including foregone tax revenues and social insurance contributions from workers who exited the labour force, the total fiscal burden reached an estimated 12.5 billion euros (Börsch-Supan et al. 2015). These costs are financed through current contribution rates, borne by the remaining working population.

A subsequent reform, the *Flexirentengesetz* of January 2017, introduced more flexible partial retirement options and adjusted earnings limits for early retirees who wished to combine pension receipt with continued employment. To the extent that this reform altered retirement incentives for workers eligible under *Rente mit 63*, it represents a potential confound in the post-2016 period. We address this concern through a dedicated sensitivity analysis in Appendix H.

3 Data & Empirical Strategy

We use longitudinal data from the German Socio-Economic Panel (SOEP) and a difference-in-differences framework to estimate the causal effect of *Rente mit 63* on the employment of eligible men.

3.1 Data Source and Sample Construction

We use data from the German Socio-Economic Panel (SOEP), a representative longitudinal household survey conducted annually since 1984. The SOEP provides detailed information on labour market status, employment biography, household income, educational attainment, health, and regional identifiers at the individual level, making it well suited for the analysis of retirement behaviour.

Our outcome variable is labour force participation (LFP), coded as a binary indicator equal to one if the respondent reports being employed full-time or part-time (SOEP variable `plb0022_h` categories 1 and 2) and zero for all other labour market states, including registered unemployment, retirement, and non-participation (categories 3 through 11). The key assignment variable is contribution years, which we approximate as full-time work experience plus one half of part-time work experience, using the generated SOEP variables `pgexpft` and `pgexppt`. This weighting mirrors the partial crediting of part-time employment periods under German pension law, though it may deviate from official contribution year counts for workers with complex employment histories; we discuss this measurement concern in Section 6.3.

The treatment group comprises men with at least 45 contribution years, who became eligible for penalty-free early retirement under *Rente mit 63*. The control group comprises men with 42 to 44 contribution years, who faced identical labour market conditions and age-related employment incentives but fell short of the eligibility threshold. We restrict the sample to men aged 62 to 65, the age window in which the reform's early retirement incentives are most salient. The observation period spans 2009 to 2019, with the reform year 2014 excluded to avoid contamination from partial treatment exposure during the transition period. The pre-reform period thus covers 2009 to 2013, and the post-reform period covers 2015 to 2019.

The resulting estimation sample contains 2,068 person-year observations from 838 unique individuals, of whom 572 observations belong to the treatment group and 1,496 to the control group. Table 3.1 reports summary statistics for key variables, separately by treatment status.

Table 3.1: Summary Statistics by Treatment Status and Period

	Treatment (≥45)		Control (42 - 44)	
	Pre (2009 - 2013)	Post (2015 - 2019)	Pre (2009 - 2013)	Post (2015 - 2019)
Observations	283	289	659	837
Age (years)	63.9	64.1	63.4	63.5
Contribution years	46.4	46.5	43.2	43.2
Household income (€)	38,815	51,692	39,186	46,751
Education (years)	10.5	10.7	11.2	11.3
Employment rate (%)	62.5	59.2	51.6	61.2
East Germany (%)	11.0	20.1	16.7	26.2
Physical health (PCS)	46.6	45.2	46.6	46.2
Mental health (MCS)	54.2	55.6	53.5	54.2

Notes: Source: SOEP. Household income: equivalised post-government income (pequiv i11102). Contribution years = full-time experience + 0.5 × part-time experience. PCS/MCS: SF-12 physical/mental component scores (higher = better health). Sample: men aged 62-65, 2009 - 2019 (excl. 2014).

3.2 Difference-in-Differences Framework

Our identification strategy exploits the discrete eligibility threshold at 45 contribution years in a difference-in-differences (DiD) design. The baseline specification estimates the pooled OLS model:

$$\text{LFP}_{it} = \beta_0 + \beta_1 \text{Treat}_i + \beta_2 \text{Post}_t + \beta_3 (\text{Treat}_i \times \text{Post}_t) + \varepsilon_{it} \tag{1}$$

where Treat_i equals one for individuals with at least 45 contribution years, Post_t equals one for observations in 2015 to 2019, and β_3 is the DiD estimand of interest, capturing the causal effect of eligibility on employment under the parallel trends assumption. Standard errors are clustered at the individual level throughout to account for serial correlation within person.

Our preferred specification replaces the pooled intercept with individual fixed effects:

$$\text{LFP}_{it} = \alpha_i + \beta_1 \text{Post}_t + \beta_2 (\text{Treat}_i \times \text{Post}_t) + \varepsilon_{it} \tag{2}$$

where α_i absorbs all time-invariant individual heterogeneity, including factors such as lifetime earnings capacity, occupational sector, and baseline health endowments that may simultaneously predict treatment status and employment outcomes. The Treat_i indicator is collinear with the individual fixed effects and drops from the model. This specification exploits only within-person variation over time and constitutes Model (4) in the results. We additionally report two intermediate specifications: Model (1) without the interaction term as a descriptive baseline, and Model (3) with year fixed effects in place of the Post_t indicator.

3.3 Control Group Selection and Parallel Trends

The validity of the DiD design rests on the assumption that treatment and control group would have followed parallel employment trajectories in the absence of the reform. The choice of control group involves a trade-off between comparability and sample size. A narrow control group (e.g. 44 contribution years only) maximises similarity to the treatment group but yields too few observations for reliable inference. A broad control group (e.g. 35 to 44 years) provides statistical power but may include workers whose labour market attachment differs systematically from the treatment group.

We evaluate four candidate control groups with lower bounds at 35, 40, 41, and 42 contribution years. For each specification, we estimate a pre-reform event-study model with individual and year fixed effects and test the joint significance of all pre-treatment interaction coefficients (2009 to 2012, reference year 2013) using a chi-squared test. Table B.1 in Appendix B reports the results. All four specifications fail to reject the null hypothesis of zero pre-treatment coefficients at the 10 percent significance level, providing support for the parallel trends assumption across all candidate definitions.

We select the 42 to 44 specification as our main control group on the grounds of comparability. Workers with 42 to 44 contribution years are closest to the treatment group in terms of accumulated labour market experience and, by extension, in the observable characteristics that shape both employment trajectories and retirement decisions. Broader control groups include workers with substantially shorter employment histories, who may differ systematically in occupational composition, earnings profiles, and health endowments, raising the risk that the parallel trends assumption holds only conditionally. Figure B.1 in Appendix B illustrates this pattern: the pre-reform employment level of the 42 to 44 control group lies closest to that of the treatment group, whereas the broader specifications show progressively larger level gaps, consistent with the inclusion of workers with weaker labour market attachment. The control-to-treatment ratio of 2.6:1 provides sufficient statistical power for reliable inference. The pooled OLS DiD estimates for all four control group definitions are reported in Table C.1 in Appendix C; the coefficient remains negative and statistically significant across all specifications, confirming that the main finding does not depend on the choice of a particular threshold.

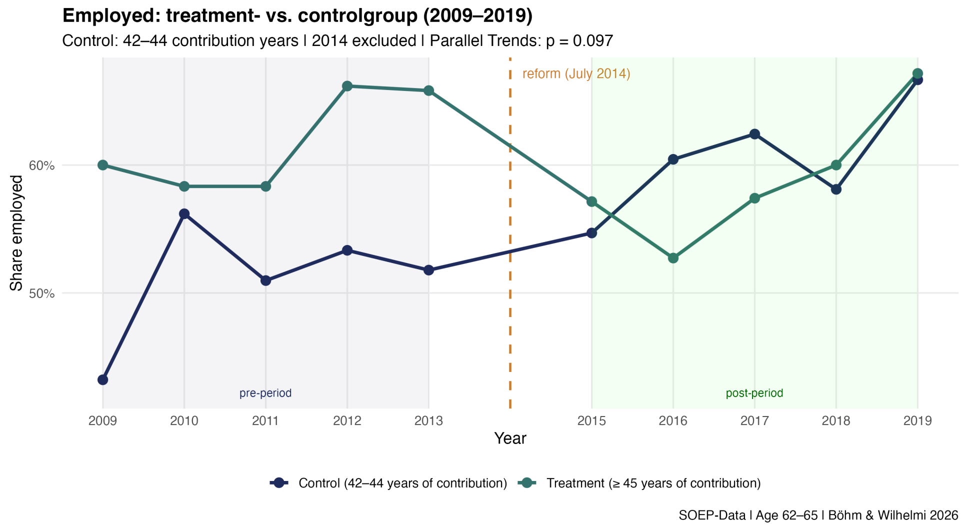

Figure 3.1: Employment rates for treatment (≥45 contribution years) vs. control group (42-44 contribution years), 2009 - 2019. Dashed line marks the introduction of *Rente mit 63* in July 2014. Year 2014 excluded from estimation sample. Joint parallel trends test: $\chi^2 = 7.868$, $df = 4$, $p = 0.097$. Source: SOEP. Sample: men aged 62-65.

3.4 Heterogeneity Specifications

To investigate whether the reform effect varies along observable dimensions, we augment the individual fixed effects model with triple interaction terms:

$$\text{LFP}_{it} = \alpha_i + \beta_1 \text{Post}_t + \beta_2(\text{Treat}_i \times \text{Post}_t) + \beta_3(\text{Treat}_i \times \text{Post}_t \times X_i) + \varepsilon_{it} \tag{3}$$

where X_i is a binary moderator variable. In this specification, β_2 captures the treatment effect for the reference group ($X_i = 0$), while β_3 measures the differential effect for the group defined by $X_i = 1$. We estimate three variants. For regional heterogeneity, X_i is a binary East Germany indicator (federal states of Brandenburg, Mecklenburg-Vorpommern, Saxony, Saxony-Anhalt, and Thuringia). For income heterogeneity, X_i indicates above-median equivalised household income. For educational heterogeneity, X_i indicates above-median years of schooling.

As an extension to the binary income split, we estimate a quintile-level specification that interacts the DiD term with dummies for income quintiles Q2 through Q5, with Q1 (the lowest quintile) as the reference category. This finer partition allows us to trace the treatment effect across the full income distribution rather than imposing a single median split.

4 Results

The results consistently point to a large and statistically significant reduction in labour force participation among eligible men following the introduction of *Rente mit 63*. The main effect is robust across model specifications and control group definitions. Heterogeneity analyses reveal no significant differences by region or educational attainment, while the income distribution shows a noteworthy gradient concentrated at the lower end of the distribution.

4.1 A Large and Robust Employment Reduction

Table 4.1 presents the results of four progressively richer model specifications, all estimated on a sample of 2,068 person-year observations covering men aged 62 to 65 over the period 2009 to 2019, excluding the reform year 2014. The stepwise structure allows us to assess the sensitivity of the DiD estimate to the successive inclusion of time and individual fixed effects.

Table 4.1: Main Results - Effect of *Rente mit 63* on Labour Force Participation

	Model (1)	Model (2)	Model (3)	Model (4)
Variables				
Treat	0.042	0.110^{*}	0.110^{*}	
	(0.029)	(0.043)	(0.043)	
Post	0.060^{*}	0.096^{**}		-0.342^{***}
	(0.028)	(0.032)		(0.057)
Treat × Post		-0.130^{*}	-0.125^{*}	-0.219^{***}
		(0.058)	(0.058)	(0.050)
Observations	2,068	2,068	2,068	1,674
Year FE	No	No	Yes	No
Individual FE	No	No	No	Yes

Notes: Clustered standard errors at the individual level in parentheses. Control group: 42 - 44 contribution years. Sample: men aged 62 - 65, 2009 - 2019 (excl. 2014). Models (1) - (3) use all 2,068 person-year observations. Model (4) uses 1,674 observations after removal of 394 singleton observations without within-person variation, as required by the individual fixed effects estimator. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Model (1) serves as a descriptive baseline without a DiD interaction term, capturing only average differences in employment between treatment and control group across time. Model (2) introduces the DiD interaction term as specified in Equation 1. The estimated coefficient on Treat × Post is -0.130 ($p < 0.05$), implying that eligibility for early retirement under *Rente mit 63* reduced the probability of employment by approximately 13 percentage points relative to the control group. Model (3) augments this specification with year fixed effects to absorb common aggregate time trends, yielding a comparable estimate of -0.125 ($p < 0.05$).

Our preferred specification, Model (4), corresponds to Equation 2 and replaces the common time trend with individual fixed effects, exploiting only within-person variation over time. This absorbs all time-invariant unobserved heterogeneity at the person level, including factors such as lifetime earnings capacity, occupational sector, and baseline health endowments that may simultaneously predict treatment status and employment outcomes. The resulting DiD estimate is -0.219 ($p < 0.001$), implying a reduction in the probability of being employed of approximately 22 percentage points among eligible men. To contextualise this magnitude: the pre-reform employment rate in the treatment group stood at 62.5 percent, implying that the probability of continued employment fell by approximately 35 percent relative to the pre-reform baseline.

The larger magnitude of Model (4) relative to the pooled OLS specifications is consistent with positive selection into the treatment group. Workers who accumulate at least 45 contribution years tend to have above-average employment propensities, so that cross-sectional comparisons understate the causal effect. Once individual fixed effects remove this time-invariant component, the reform-induced employment reduction is estimated to be stronger. The individual fixed effects estimator requires within-person variation over time and therefore drops 394 singleton observations from the estimation sample. While this mechanical sample reduction is standard practice, we note that the retained sample consists of individuals observed in both the pre- and post-reform period, which may differ compositionally from those observed in only one period; this should be borne in mind when comparing magnitudes across specifications.

The dynamic DiD estimates, plotted in Figure D.1 in Appendix D, provide additional support for the parallel trends assumption and shed light on the timing of the effect. The pre-treatment interaction coefficients for 2009 to 2012, estimated relative to the reference year 2013, are individually and jointly statistically indistinguishable from zero, lending credibility to the identifying assumption. From 2015 onwards, the treatment effect turns negative, with 2016 representing the only post-period year in which the estimate reaches conventional statistical significance ($p < 0.05$). The remaining post-period coefficients are negative but individually insignificant. While this pattern is consistent with a persistent though imprecisely estimated labour market withdrawal, the year-by-year estimates do not individually establish a sustained effect, and should be interpreted as suggestive rather than conclusive evidence on the post-reform dynamics.

4.2 Broad Uniformity Across Regions and Education

To investigate potential heterogeneity along regional and educational dimensions, we augment our preferred specification, Model (4), with interaction terms as described in Equation 3. Full regression results for all heterogeneity models are reported in Table E.1 in Appendix E. Note that the baseline DiD coefficient in the interaction models (-0.173 for RQ1, -0.329 for RQ2a, -0.212 for RQ2b) differs from the main estimate of -0.219 in Model (4), as it now captures the treatment effect for the reference group only (East = 0, HighIncome = 0, or HighEduc = 0 respectively), rather than the average effect across all observations.

For regional heterogeneity (RQ1), we interact the DiD term with a binary East Germany indicator. East and West Germany differ substantially in their labour market histories and pension biographies. In particular, workers in the former GDR were more likely to have accumulated 45 contribution years through decades of full employment under the socialist system, which could plausibly generate a stronger treatment effect in the East. The estimated coefficient on Treat $\times$ Post $\times$ East is -0.172 ($p = 0.123$), statistically indistinguishable from zero. The baseline DiD estimate remains stable at -0.173 ($p < 0.001$). We find no statistically significant difference in the reform effect between East and West German workers, though the limited statistical power of the heterogeneity analysis, as discussed in Section 6.3, means this result should not be interpreted as evidence of symmetry.

For educational heterogeneity (RQ2b), we interact the DiD term with an above-median years-of-schooling indicator. Higher educational attainment is often associated with stronger intrinsic work motivation and higher returns to continued employment, which could plausibly moderate the reform's impact. The estimated coefficient on Treat $\times$ Post $\times$ HighEduc is -0.016 ($p = 0.868$), statistically indistinguishable from zero. This absence of a detectable education gradient is consistent with the reform's design: eligibility depends entirely on contribution years rather than qualification levels, so workers across the educational spectrum who meet the threshold face identical financial incentives to exit the labour market. Given the limited sample size, however, we cannot rule out that a meaningful education gradient exists but remains undetected at conventional significance levels. The income distribution, however, tells a different story.

4.3 An Income Gradient: Lower Earners Exit More Strongly

To examine potential heterogeneity along the income distribution, we estimate two complementary specifications. First, a binary split at the median household income yields an interaction coefficient on Treat $\times$ Post $\times$ HighIncome of $+0.160$ ($p = 0.062$), which falls just short of conventional significance levels. This suggests that above-median earners experience a somewhat weaker employment decline, but the binary split lacks the resolution to capture the full distributional pattern.

We therefore turn to a quintile-level specification, interacting the DiD term with dummies for income quintiles Q2 through Q5, with Q1 as the reference category. The full regression output and pairwise comparison tests are reported in Appendix E. Figure 4.1 plots the implied absolute treatment effects for each quintile, calculated as linear combinations of the baseline DiD coefficient and the relevant quintile interaction term.

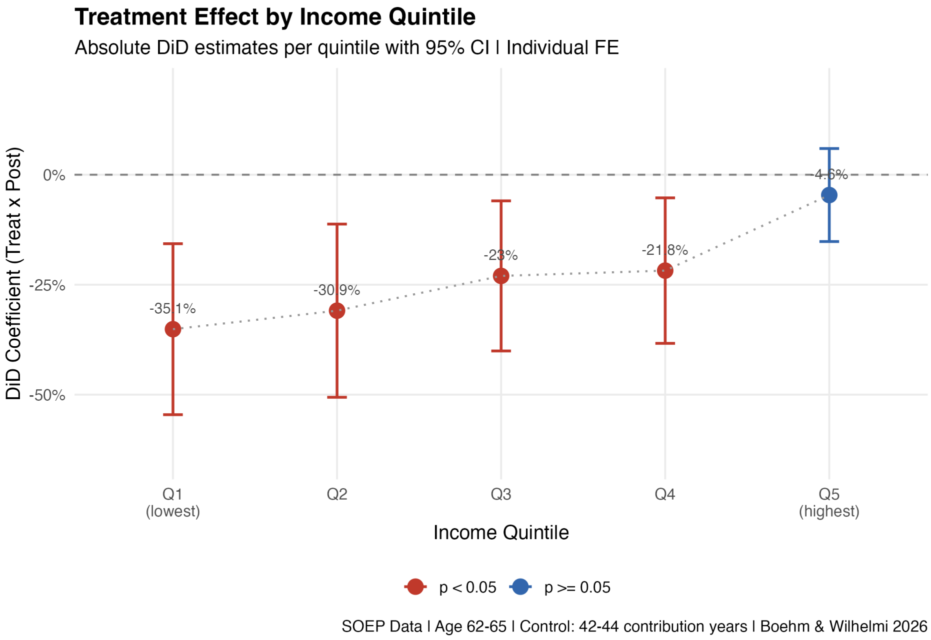

Figure 4.1: DiD Treatment Effects by Income Quintile (Individual FE).

The estimates reveal a pattern suggestive of a gradient across the income distribution, with the employment decline concentrated among lower earners and attenuating toward the top of the distribution. The absolute treatment effects are statistically significant for all four lower quintiles, ranging from -0.351 ($p < 0.001$) for Q1 to -0.218 ($p < 0.05$) for Q4. The top quintile alone yields an absolute effect of -0.046 ($p = 0.392$), statistically indistinguishable from zero. A direct test of the difference between Q1 and Q5 confirms that the two extreme quintiles differ by approximately 30.5 percentage points ($t = -2.782$, $p < 0.01$); full pairwise comparison results are reported in Appendix E.

Bonferroni-corrected pairwise tests between all quintile pairs do not yield additional significant differences after correction for multiple comparisons. The pattern across quintiles should therefore be interpreted as suggestive evidence of a distributional gradient rather than as proof of a strictly monotonic relationship, given that adjacent quintile differences are not individually significant. The reform-induced employment reduction is nonetheless heavily concentrated among lower-income workers while leaving the labour market behaviour of top earners largely unaffected.

This pattern is consistent with the financial structure of the reform. Workers in lower income quintiles tend to face higher pension replacement rates relative to their pre-retirement earnings, making penalty-free early retirement particularly attractive. Higher earners, by contrast, face lower replacement rates and may have stronger financial incentives to remain in employment.

5 Robustness Checks

The main finding of a large and significant employment reduction is subject to four robustness checks: alternative control group definitions, placebo tests in false reform years, a donut design excluding workers near the eligibility threshold, and a sensitivity analysis accounting for the Flexirente reform of 2017. Across all checks, the direction and approximate magnitude of the main effect are preserved.

5.1 Stability Across Alternative Control Groups

To assess whether the main result is sensitive to the specific choice of control group lower bound, we re-estimate the baseline OLS DiD model with three alternative lower bounds: 35, 40, and 41 contribution years. The results are reported in Table C.1 in Appendix C.

Across all four specifications, the DiD coefficient remains negative and statistically significant, ranging from -0.170 ($p < 0.01$) for the 35-44 control group to -0.130 ($p < 0.05$) for the main 42-44 specification. The stability of the result across control group definitions with vastly different control-to-treatment ratios, ranging from 10.5:1 for the 35-44 group to 2.6:1 for the main specification, confirms that the finding is not an artefact of the particular threshold chosen.

5.2 No Evidence of Pre-Trends: Placebo Tests

To test for the absence of pre-existing trends, we conduct placebo tests by simulating the reform in three false years: 2010, 2011, and 2012. In each test, we construct a fake post indicator using only pre-reform observations (2009 - 2013) and re-estimate the individual fixed effects model. The results are plotted in Figure F.1 in Appendix F.

The placebo estimates for 2011 and 2012 are statistically indistinguishable from zero (-0.072, $p = 0.239$ and $+0.059$, $p = 0.307$ respectively), consistent with the absence of spurious pre-trends. The 2010 placebo yields a significant estimate of -0.144 ($p < 0.01$), which warrants careful interpretation. This test uses only a single pre-period year (2009), which provides insufficient variation for the individual fixed effects estimator to reliably absorb person-level heterogeneity; the resulting estimate is therefore mechanically less stable than those for 2011 and 2012. We nonetheless acknowledge that a significant placebo estimate cannot be dismissed entirely on purely technical grounds. Taken together, the three placebo tests provide mixed but overall reassuring evidence: the two better-identified tests yield null results, while the 2010 result reflects a structural limitation of the test design rather than compelling evidence of differential pre-trends. The parallel trends assumption is further supported by the formal chi-squared test reported in Table B.1, which fails to reject the null of zero pre-treatment coefficients across all four pre-reform years jointly ($p = 0.097$).

5.3 No Bunching at the Eligibility Threshold

A potential concern is that workers near the 45-year eligibility threshold may have manipulated their contribution year counts in anticipation of the reform, biasing the DiD estimate. To address this, we apply a donut robustness design, excluding all observations with contribution years in the range $[44, 45)$ from the estimation sample. This removes 417 person-year observations from the raw sample, reducing it from 2,068 to 1,651 observations. After additionally removing singleton observations without within-person variation, as in the main specification, the final donut estimation sample comprises 1,232 observations across 503 individuals, compared to 1,674 in the main specification.

The individual fixed effects estimate on the donut sample is -0.194 ($p < 0.01$), remaining negative, statistically significant, and of comparable magnitude to the main specification of -0.219. The stability of the estimate confirms that the main result is not driven by sorting near the eligibility threshold. The heterogeneity specifications lose statistical significance in the donut sample (Figure G.1), which is expected given the additional reduction in sample size; this reinforces the limited power caveat noted for the heterogeneity analyses in Section 6.3.

5.4 Flexirente Does Not Drive the Effect

A potential confound in the post-period is the Flexirente reform of January 2017, which introduced more flexible partial retirement options and adjusted earnings limits for early retirees who wished to continue working. If the Flexirente systematically altered retirement incentives for the treatment group, it could confound our post-2016 estimates.

We conduct two sensitivity checks. First, we augment Model (4) with a binary post-2017 indicator (Flexirente dummy). The DiD estimate remains negative and significant at -0.114 ($p < 0.05$), while the Flexirente dummy itself is large and significant at -0.342 ($p < 0.001$), capturing a general post-2017 employment decline affecting both groups equally. Crucially, this specification holds the common post-2017 trend constant and continues to identify a negative treatment-control differential, providing direct evidence that a *Rente mit 63* signal persists beyond 2016 after accounting for any symmetric Flexirente influence. Second, we restrict the post-period to 2015 - 2016 only, thereby excluding all post-Flexirente observations. The DiD estimate is -0.088 ($p = 0.267$), losing conventional statistical significance. Two competing explanations account for this result. The substantially smaller post-period sample (1,072 vs. 1,674 observations) reduces statistical power considerably. At the same time, the point estimate itself declines from -0.219 to -0.088, a reduction of approximately 60 percent that cannot be attributed to reduced statistical power alone and reflects genuine uncertainty about the magnitude of the effect in the earlier post-reform years. Full results are reported in Appendix H.

We retain the full 2015 - 2019 post-period as our main specification for two reasons. First, the Flexirente dummy check demonstrates that a significant treatment-control differential survives after the common post-2017 employment trend is partialled out, suggesting that the full-period estimate is not driven by a Flexirente confound operating symmetrically across groups. Second, restricting the post-period to 2015 - 2016 yields too few observations for the individual fixed effects estimator to produce reliable inference. We therefore interpret -0.219 as the best available estimate of the average treatment effect over the full post-reform period 2015 - 2019, while acknowledging that the true effect may be concentrated in the earlier post-reform years. The extent to which the estimate reflects *Rente mit 63* versus Flexirente influences in the 2017 - 2019 window cannot be fully resolved with the available data; this is discussed as a limitation in Section 6.3.

6 Discussion

The results establish a large and robust employment effect of *Rente mit 63*, with no statistically significant heterogeneity detected by region or educational attainment, and a pronounced income gradient concentrated among lower earners. We now discuss the interpretation and implications of these findings before turning to the limitations of our analysis.

6.1 Interpreting the Magnitude of the Main Effect

The individual fixed effects estimate of -0.219 ($p < 0.001$) implies a reduction in employment probability of approximately 22 percentage points among eligible men, with the full post-reform period required to achieve precise identification. Crucially, this reflects a genuine withdrawal from employment rather than a shift in pension claiming timing alone: Dolls and Krolage (2023) show using administrative pension records that eligible men claim their pensions on average nine months earlier than comparable non-eligible workers, confirming that the reduction in measured employment corresponds to actual labour market exit. The larger magnitude of the individual fixed effects estimate relative to pooled OLS is consistent with positive selection into the treatment group: workers who accumulate at least 45 contribution years tend to have above-average employment propensities, so cross-sectional comparisons understate the causal effect. The magnitude falls within the range documented in the broader literature on pension reform and labour supply (Riphahn and Schrader 2023; Engels et al. 2017), lending further credibility to the view that financial retirement incentives are a powerful lever for older workers' employment decisions.

6.2 Why Lower Earners Respond More Strongly

The income gradient documented in Section 4.3 raises the question of what mechanisms drive the differential response across the income distribution. We discuss three candidate explanations, while acknowledging that our data do not allow us to formally distinguish between them.

A first mechanism operates through pension replacement rates. Workers in lower income quintiles typically face higher replacement rates relative to their pre-retirement earnings, making the financial cost of early retirement comparatively low. This is consistent with evidence in Riphahn and Schrader (2023), who document that individuals with low pension wealth responded more strongly to early retirement reforms, a pattern aligned with our income gradient.

A second mechanism relates to occupational characteristics. Workers in lower income quintiles are more likely to be employed in physically demanding occupations, where the disutility of continued work is high. Among pre-reform treatment group members, 49.1 percent of workers in the lowest income quintile are classified as manual workers (blue-collar *Arbeiter*), compared to only 14.3 percent in the top quintile ($\chi^2 = 15.651$, $df = 4$, $p = 0.004$; see Table I.1). This occupational sorting is consistent with the hypothesis that *Rente mit 63* offered a particularly valued exit option for lower-income workers in physically demanding roles, though the descriptive nature of this evidence does not allow us to establish the channel causally.

A third possibility is that high earners had already accumulated sufficient private wealth to finance early retirement through alternative channels prior to the reform, while lower earners lacked the financial means to exit earlier without it (Börsch-Supan et al. 2015). Disentangling these mechanisms would require data on individual replacement rates and private wealth holdings beyond what the SOEP provides.

6.3 Limitations

Several limitations warrant acknowledgement. First, our sample is restricted to men. Women are structurally less likely to meet the 45-year contribution threshold due to career interruptions related to child-rearing, and Geyer and Welteke (2021) document that women respond differently to retirement age reforms, with passive rather than active programme substitution dominating. Whether *Rente mit 63* generated comparable patterns for the smaller subset of eligible female workers remains an important open question.

Second, the treatment group comprises only 572 person-year observations, which constrains the statistical power of the heterogeneity analyses. The absence of significant heterogeneity by region and education should therefore be interpreted cautiously, as it may reflect limited power rather than genuine uniformity.

Third, our contribution year approximation via SOEP experience variables introduces potential measurement error in treatment assignment, as it may deviate from official Deutsche Rentenversicherung counts for workers with complex histories. Under standard assumptions, this generates attenuation bias, implying that -0.219 is likely a lower bound on the true effect.

Fourth, the estimate of -0.219 reflects the average treatment effect over the full post-reform period 2015 - 2019 and should be interpreted accordingly. When the post-period is restricted to 2015 - 2016, the coefficient shrinks to -0.088, a reduction of approximately 60 percent that reflects both reduced statistical power and genuine uncertainty about the effect magnitude in the earlier post-reform years. While the Flexirente dummy check in Section 5.4 demonstrates that a significant treatment-control differential persists after the common post-2017 employment trend is partialled out, the contribution of *Rente mit 63* versus Flexirente influences to the full-period estimate in the 2017 - 2019 window cannot be resolved with the available data. Researchers requiring a conservative lower bound on the *Rente mit 63* effect should therefore place greater weight on the 2015 - 2016 restricted estimate, bearing in mind its limited statistical power.

Finally, SOEP wealth data fall outside our panel window in sufficient density, making wealth variables incompatible with the individual fixed effects approach and precluding a direct test of the private wealth mechanism discussed above.

7 Conclusion

This paper estimates the causal effect of Germany's *Rente mit 63* on the labour force participation of eligible men aged 62 to 65. Using SOEP data and a difference-in-differences design that compares workers with at least 45 contribution years to a closely matched control group with 42 to 44 contribution years, we find that the reform reduced the probability of employment by approximately 22 percentage points ($p < 0.001$). This estimate is robust across alternative control group definitions, placebo tests in false reform years, and a donut design excluding workers near the eligibility threshold.

The reform effect is broadly uniform across regions and educational groups. The income distribution, however, reveals a pronounced gradient: the employment decline ranges from 35.1 percentage points in the lowest income quintile to a statistically insignificant 4.6 percentage points in the highest. This pattern is consistent with higher pension replacement rates and greater physical job demands among lower earners, both of which make penalty-free early retirement particularly attractive for this group.

These findings speak directly to the fiscal sustainability of the implicit generational contract that underlies pay-as-you-go pension systems. The behavioural response documented here is not merely a shift in claiming timing, as shown by Dolls and Krolage (2023), but a genuine reduction in employment. Each eligible worker who exits the labour force generates a triple fiscal cost: foregone income tax, foregone social insurance contributions, and additional years of benefit receipt. At the scale documented by Börsch-Supan et al. (2015), with approximately 240,000 eligible workers per cohort, these costs accumulate rapidly. The income gradient sharpens the distributional question raised at the outset: the costs of the reform are socialised across all contributors through higher contribution rates, while the behavioural response is concentrated among lower-income workers who exit the labour force in large numbers, and largely absent among top earners who continue working regardless. The reform thus does not create new resources; it redistributes existing ones from those who continue to pay in to those who leave, enacted at a time when Germany's shrinking working-age population makes every additional year of labour force participation increasingly valuable.

Our analysis is subject to several limitations discussed in Section 6.3, including a relatively small treatment group that constrains the power of heterogeneity tests, potential measurement error in the contribution year approximation, and the inability to fully disentangle the *Rente mit 63* effect from the Flexirente reform of 2017 in the later post-period.

Two directions for future research emerge from our findings. First, an analogous analysis for women would test whether the income gradient extends to the smaller subset of female workers who meet the 45-year threshold, a group whose retirement behaviour may differ fundamentally given the role of career interruptions documented by Geyer and Welteke (2021). Second, linking our survey-based evidence with administrative pension records would allow researchers to observe exact contribution histories, eliminating the measurement error inherent in our SOEP approximation, and to examine the reform's effects on pension wealth accumulation and lifetime fiscal balances.

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A Birth and Death Statistics in Germany

Figure A.1 displays the annual number of live births and deaths in Germany from 2000 to 2025. The shaded area represents the natural population deficit, defined as the excess of deaths over births. While Germany has recorded more deaths than births throughout the entire period, the deficit has widened considerably in recent years. In 2025, Germany recorded just 654,300 births, the lowest figure since 1946, while deaths exceeded births by approximately 352,000, the largest post-war demographic deficit on record (Statistisches Bundesamt 2026). The vertical line marks the introduction of *Rente mit 63* in July 2014, illustrating that the reform took effect against a backdrop of already deteriorating demographic fundamentals.

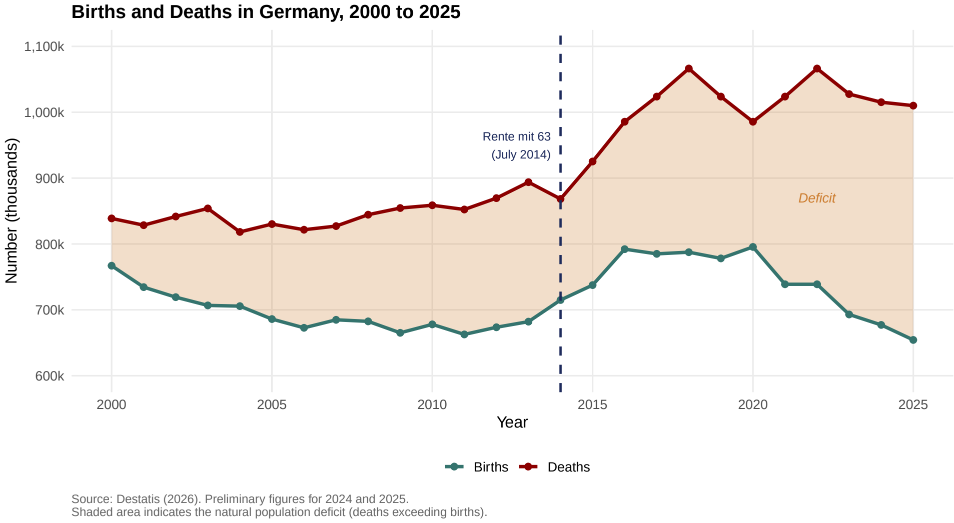


Figure A.1: Annual number of births and deaths in Germany, 2000 to 2025. The shaded area indicates the natural population deficit (deaths exceeding births). The vertical line marks the introduction of *Rente mit 63* in July 2014. Figures for 2024 and 2025 are preliminary.

Figure A.2 complements this picture by tracing the old-age dependency ratio (Altenquotient) from 1950 to 2024, with projections to 2070. The ratio measures the number of persons aged 65 and older per 100 persons of working age (20 to 64). In 1950 it stood at 16; by 2024 it had risen to 39. The 16th coordinated population projection of Destatis projects a further steep increase to between 43 and 61 by 2070, depending on assumptions about fertility, life expectancy, and net migration (Statistisches Bundesamt 2025). The shaded band represents this projection range. The vertical line again marks July 2014, underscoring that *Rente mit 63* was enacted at the onset of the steepest projected phase of demographic ageing in post-war German history.

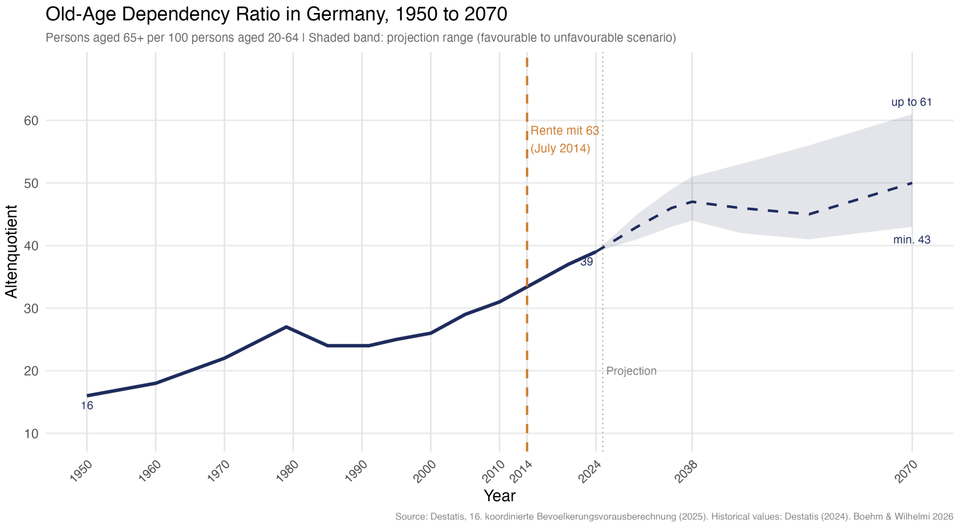


Figure A.2: Old-age dependency ratio (Altenquotient) in Germany, 1950 to 2070. The ratio measures the number of persons aged 65 and older per 100 persons aged 20 to 64. The solid line shows historical values; the dashed line and shaded band show the projection range from the 16th coordinated population projection (Statistisches Bundesamt 2025), spanning the most favourable scenario (43 in 2070) to the least favourable (61 in 2070). The vertical line marks the introduction of *Rente mit 63* in July 2014.

B Parallel Trends Tests

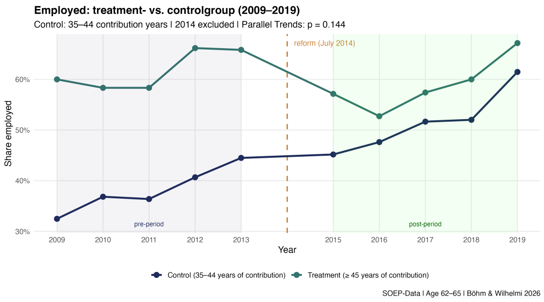
Table B.1 reports the joint parallel trends test results for all four candidate control group specifications. For each specification, we estimate a pre-reform event-study model with individual and year fixed effects and test the joint significance of all pre-treatment interaction coefficients (2009 - 2012, reference year: 2013) using a chi-squared test. The null hypothesis is that all pre-treatment coefficients equal zero. All four specifications fail to reject the null hypothesis at the 10% significance level, providing support for the parallel trends assumption underlying our DiD design. The 42 - 44 specification shows no evidence of pre-existing divergence ($p = 0.097$) and most closely tracks the treatment group visually in the pre-reform period, as shown in Figure B.1.

Table B.1: Joint Parallel Trends Test - Pre-Reform Period 2009 - 2012

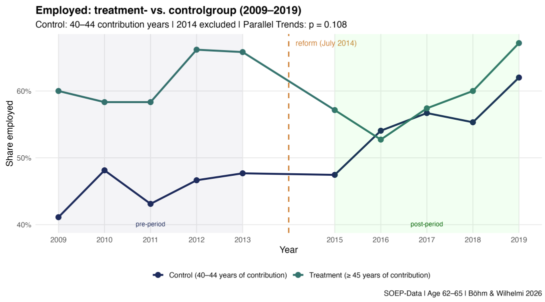
Control Group	N (Control)	Chi^2	df	p-value
35 - 44	5,981	6.847	4	0.144
40 - 44	2,866	7.593	4	0.108
41 - 44	2,159	8.014	4	0.091
42 - 44*	1,496	7.868	4	0.097

Notes: H_0: All pre-treatment interaction coefficients equal zero. Clustered standard errors at the individual level. Treatment group: $\geq$45 contribution years (N = 572 across all specifications). * Main specification.

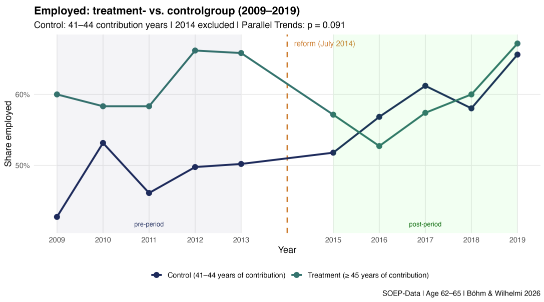
Figure B.1 presents the raw employment trends for all four control group specifications side by side. All panels show broadly parallel pre-reform trajectories, consistent with the formal test results in Table B.1. The 42 - 44 specification most closely tracks the treatment group in the pre-period and serves as the main specification throughout the paper.



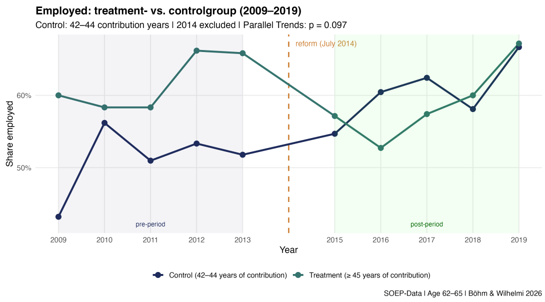
(a) Control: 35 - 44 contribution years



(b) Control: 40 - 44 contribution years



(c) Control: 41 - 44 contribution years



(d) Control: 42 - 44 contribution years (main)

Figure B.1: Employment rates for treatment (≥45 contribution years) vs. control groups across all four specifications, 2009 - 2019. Dashed line marks the introduction of *Rente mit 63* in July 2014. Year 2014 excluded from estimation sample. Source: SOEP. Sample: men aged 62 - 65.

C Alternative Control Groups

Table C.1 reports the pooled OLS DiD estimates for all four control group specifications. The DiD coefficient remains negative and statistically significant across all specifications, confirming that the main finding does not depend on the choice of a particular threshold. The 42 - 44 specification serves as the main control group throughout the paper, as discussed in Section 3.3.

Table C.1: Robustness - Alternative Control Group Definitions (OLS DiD)

	35 - 44	40 - 44	41 - 44	42 - 44*
Variables				
Treat	0.239***	0.170***	0.139***	0.110*
	(0.038)	(0.040)	(0.041)	(0.043)
Post	0.136***	0.103***	0.108***	0.096**
	(0.018)	(0.025)	(0.028)	(0.032)
Treat × Post	−0.170***	−0.137*	−0.142*	−0.130*
	(0.053)	(0.055)	(0.056)	(0.058)
N (Control)	5,981	2,866	2,159	1,496
N (Total)	6,553	3,438	2,731	2,068
Ratio (C:T)	10.5:1	5.0:1	3.8:1	2.6:1

Notes: Clustered standard errors at the individual level in parentheses. Treatment group: ≥45 contribution years (N = 572 across all specifications). * Main specification. Sample: men aged 62 - 65, 2009 - 2019 (excl. 2014). $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

D Dynamic DiD - Event Study

Figure D.1 presents the dynamic difference-in-differences estimates, obtained by interacting the treatment indicator with year dummies relative to the reference year 2013, with year fixed effects. The pre-treatment coefficients for 2009 to 2012 are individually and jointly statistically indistinguishable from zero, corroborating the parallel trends assumption. In the post-reform period, the treatment effect turns negative from 2015 onwards, with 2016 representing the only year in which the estimate reaches conventional statistical significance. The remaining post-period coefficients are negative but individually insignificant, consistent with a sustained though imprecisely estimated labour market withdrawal among eligible workers.

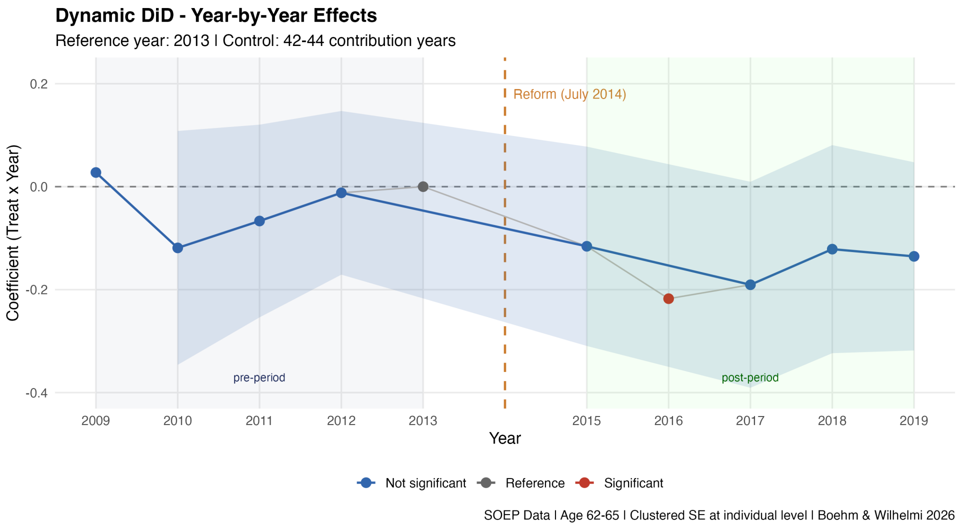


Figure D.1: Dynamic DiD - Year-by-Year Treatment Effects

E Heterogeneity Regression Results

Table E.1 reports the full regression results for the regional and educational heterogeneity specifications discussed in Section 4.2. All models include individual fixed effects and cluster standard errors at the individual level. None of the interaction terms reach conventional levels of statistical significance, consistent with the discussion in the main text.

Table E.1: Heterogeneity Analysis - Individual Fixed Effects

	RQ1: East/West	RQ2a: Income	RQ2b: Education
Variables			
Post	-0.347^{***}	-0.346^{***}	-0.342^{***}
	(0.056)	(0.056)	(0.057)
Treat × Post	-0.173^{***}	-0.329^{***}	-0.212^{***}
	(0.052)	(0.078)	(0.058)
Treat × Post × East	-0.172		
	(0.111)		
Treat × Post × High Income		$+0.160$	
		(0.085)	
Treat × Post × High Educ			-0.016
			(0.094)
Observations	1,674	1,674	1,674
Individual FE	Yes	Yes	Yes

Notes: Clustered standard errors at the individual level in parentheses. Control group: 42 - 44 contribution years. Sample: men aged 62 - 65, 2009 - 2019 (excl. 2014). $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table E.2 reports the full quintile-level regression results underlying Figure 4.1 in the main text. The baseline DiD coefficient captures the effect for the lowest income quintile (Q1), while the interaction terms measure the differential effect of each quintile relative to Q1.

Table E.2: Income Quintile Analysis - Individual Fixed Effects

	DiD Estimate
Variables	
Post	-0.337^{***}
	(0.057)
Treat × Post (Q1 Ref.)	-0.351^{***}
	(0.099)
Treat × Post × Q2	+0.042
	(0.114)
Treat × Post × Q3	+0.121
	(0.123)
Treat × Post × Q4	+0.133
	(0.124)
Treat × Post × Q5	$+0.305^{**}$
	(0.110)
Observations	1,674
Individual FE	Yes

Notes: Clustered standard errors at the individual level in parentheses. Reference category: Q1 (lowest income quintile). Control group: 42 - 44 contribution years. Sample: men aged 62 - 65, 2009 - 2019 (excl. 2014). $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table E.3 reports pairwise comparison tests between income quintiles. The direct test between Q1 and Q5 confirms a statistically significant difference of approximately 30.5 percentage points ($t = -2.782$, $p < 0.01$). After Bonferroni correction for multiple comparisons, no additional quintile pair reaches conventional significance levels.

Table E.3: Pairwise Quintile Comparison Tests

Comparison	Difference (pp)	SE	t-statistic	p (raw)	p (Bonferroni)
Q1 vs Q5†	-30.5	10.96	-2.782	0.006	--
Q2 vs Q3	-7.9	12.13	-0.651	0.515	1.000
Q2 vs Q4	-9.1	12.20	-0.746	0.455	1.000
Q2 vs Q5	-26.3	11.01	-2.389	0.017	0.101
Q3 vs Q4	-1.2	9.82	-0.123	0.902	1.000
Q3 vs Q5	-18.4	9.41	-1.955	0.051	0.304
Q4 vs Q5	-17.2	8.69	-1.978	0.048	0.287

Notes: Differences reported in percentage points. † The Q1 vs Q5 comparison constitutes a single pre-specified hypothesis test based on $\hat{\beta}_{Q5}$ and is reported with the raw two-sided p-value only ($p = 0.006$, $t = -2.782$), without Bonferroni correction. The remaining 6 pairwise comparisons are between quintiles Q2 through Q5 ($\binom{4}{2} = 6$ pairs); Bonferroni correction is applied to these only. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

F Placebo Tests

Figure F.1 presents the placebo DiD estimates for fake reform years 2010, 2011, and 2012, estimated using individual fixed effects on pre-reform data only (2009 - 2013). The placebo estimates for 2011 and 2012 are statistically indistinguishable from zero, consistent with the absence of spurious pre-trends. The 2010 placebo yields a significant estimate, but is estimated with only a single pre-period year, which provides insufficient variation for the individual fixed effects estimator to reliably absorb person-level heterogeneity. This result should therefore not be interpreted as evidence against the parallel trends assumption.

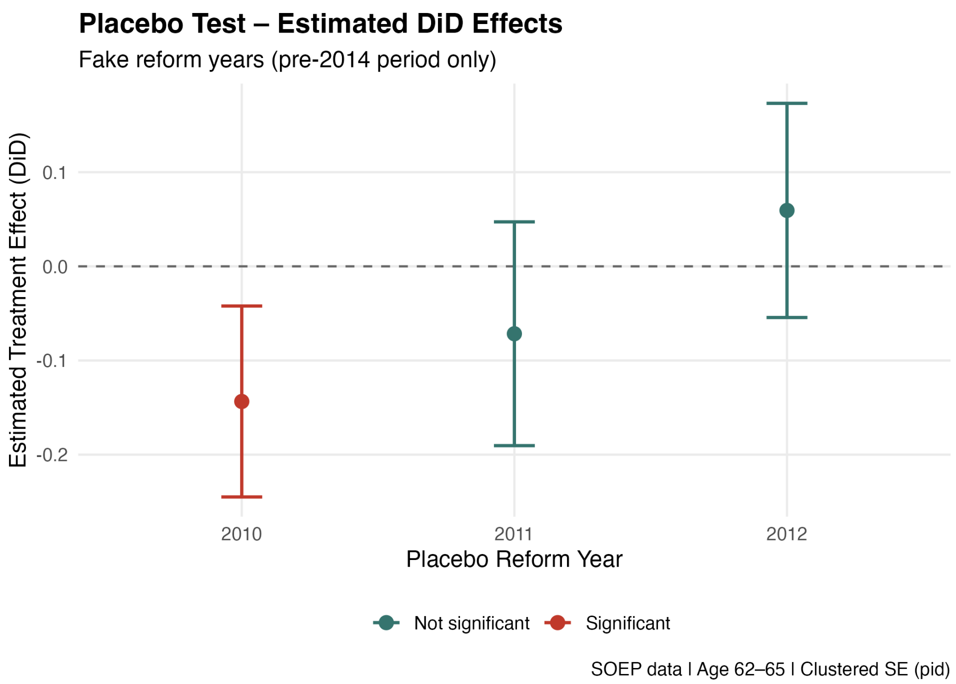


Figure F.1: Placebo DiD estimates for fake reform years 2010, 2011, and 2012. Estimated using individual fixed effects on pre-reform data only (2009 - 2013). Error bars represent 95% confidence intervals. Red: significant at 5% level; green: not significant. Control: 42 - 44 contribution years.

G Donut Robustness Design

Table G.1 compares the main individual fixed effects estimates between the full sample and the donut sample, from which all observations with contribution years in $[44, 45)$ are excluded. Figure G.1 visualises the stability of the DiD coefficient across all four model specifications. In each case, the donut estimate remains negative and of comparable magnitude to the baseline, confirming that the main result is not driven by sorting near the eligibility threshold.

Table G.1: Donut Robustness - Individual Fixed Effects (Model 4)

	Full Sample	Donut Sample (excl. contrib. years $\in [44, 45)$)
Variables		
Post	-0.342^{***}	-0.339^{***}
	(0.057)	(0.077)
Treat × Post	-0.219^{***}	-0.194^{**}
	(0.050)	(0.073)
Observations	1,674	1,232
Individual FE	Yes	Yes

Notes: Clustered standard errors at the individual level in parentheses. Control group: 42 - 44 contribution years. Sample: men aged 62 - 65, 2009 - 2019 (excl. 2014). Full sample excludes 394 singleton observations without within-person variation. Donut sample additionally excludes observations with contribution years in $[44, 45)$, reducing the sample to 1,232 observations across 503 individuals. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

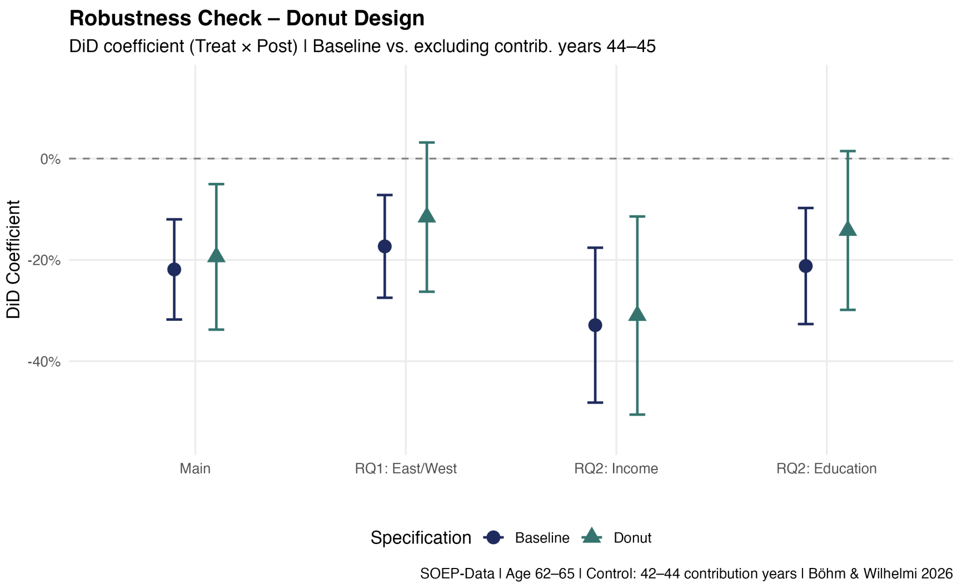


Figure G.1: Donut Robustness Check. DiD coefficients (Treat × Post) for baseline and donut specifications across all four model specifications. Error bars represent 95% confidence intervals. Control: 42 - 44 contribution years.

H Flexirente Sensitivity Analysis

Table H.1 reports the results of two sensitivity checks addressing the potential confounding effect of the Flexirente reform of January 2017. The baseline specification is Model (4) from the main text. The Flexirente dummy specification augments Model (4) with a binary indicator equal to one for observations from 2017 onwards. The restricted post-period specification limits the post-reform window to 2015 - 2016 only, thereby excluding all post-Flexirente observations.

Table H.1: Flexirente Sensitivity - Individual Fixed Effects

	Baseline	Flexi-Dummy	2015 - 16 only
Variables			
Post	-0.342^{***}	-0.396^{***}	-0.410^{***}
	(0.057)	(0.057)	(0.064)
Treat × Post	-0.219^{***}	-0.114^{*}	-0.088
	(0.050)	(0.052)	(0.079)
Post-2017 (Flexirente)		-0.342^{***}	
		(0.045)	
Observations	1,674	1,674	1,072
Individual FE	Yes	Yes	Yes
Post-Period	2015 - 19	2015 - 19	2015 - 16

Notes: Clustered standard errors at the individual level in parentheses. Control group: 42 - 44 contribution years. Sample: men aged 62 - 65, 2009 - 2019 (excl. 2014). The 2015 - 16 only specification loses statistical significance due to the substantially reduced post-period and correspondingly limited statistical power; the direction of the effect is unchanged. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

I Occupational Structure by Income Quintile

Table I.1 reports the share of manual workers (*Arbeiter*, SOEP variable `pgstib` codes 10 - 13) by income quintile among pre-reform treatment group members. Manual workers are defined as blue-collar employees in unskilled, semi-skilled, skilled, or supervisory roles. The share of manual workers declines monotonically from 49.1 percent in the lowest income quintile to 14.3 percent in the highest, consistent with the occupational mechanism discussed in Section 6.2.

Table I.1: Share of Manual Workers by Income Quintile (Pre-Reform Treatment Group)

Income Quintile	N	Manual Workers	Share (%)
Q1 (lowest)	57	28	49.1
Q2	57	20	35.1
Q3	57	19	33.3
Q4	56	19	33.9
Q5 (highest)	56	8	14.3
Total	283	94	33.2

Notes: Manual workers defined as SOEP `pgstib` codes 10 - 13 (un/angelernte Arbeiter, Facharbeiter, Vorarbeiter). Sample restricted to pre-reform period (2009 - 2013) and treatment group ($\geq$45 contribution years). N = 283 corresponds to all treatment group observations in the pre-reform period; occupational status codes are available for all observations (coverage: 100%). A Pearson chi-squared test rejects the null hypothesis of equal distribution across quintiles ($\chi^2 = 15.651$, $df = 4$, $p = 0.004$).